

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	28 June 2026
Project Name	A Comprehensive Measure of well Being
Maximum Marks	4 Marks

Step 1: Team Gathering, Collaboration and Select the Problem Statement

Select problem statement

The Human Development Index (HDI) is a composite index that measures a country's development using three key dimensions: health (life expectancy), education (mean years of schooling and expected years of schooling), and standard of living (Gross National Income per capita). Existing approaches primarily focus on predicting the HDI score or category but provide limited guidance on why a country received a particular classification or what improvements are required to achieve a higher level of human development.

Our team selected this problem because governments, policymakers, researchers, and development organizations require not only accurate HDI predictions but also actionable insights that support informed decision-making. An intelligent system capable of predicting HDI while identifying weak development indicators and recommending improvement strategies can contribute to better policy planning and resource allocation.

Problem Understanding

The current HDI prediction process answers only one question:

"What is the predicted HDI category?"

However, decision-makers also need answers to questions such as:

- Which factor is responsible for the low HDI?
- Is healthcare, education, or income affecting the country's development the most?
- What improvements are required to move from Medium HDI to High HDI?
- How can governments prioritize investments to improve human development?

The absence of such recommendations limits the usefulness of existing prediction systems.

Objective

To develop an AI-powered system that predicts the Human Development Index using key socio-economic indicators and provides personalized recommendations for improving the HDI by identifying the weakest development factors.

Step 2: Brainstorming – Idea Listing and Grouping

Idea 1: HDI Prediction

Develop a Machine Learning model to predict the HDI score and classify countries into Very High, High, Medium, or Low HDI.

Idea 2: Explain Prediction

Display the main factor (Health, Education, or Income) that most influenced the predicted HDI category.

Idea 3: Improvement Recommendation

Suggest which indicators need improvement and estimate the target values required to achieve the next HDI category.

Idea 4: Feature Importance Analysis

Show the contribution of each input feature to help users understand how the prediction was made.

Idea 5: Interactive Dashboard

Provide a dashboard displaying the HDI score, category, feature comparison, and improvement recommendations using graphs and charts.

Step 3: IDEA PRIORITIZATION

Priority	Idea	Reason
1	Improvement Recommendation	Provides actionable guidance to improve HDI and makes the project unique.
2	HDI Prediction	Core functionality required by the problem statement.
3	Explain Prediction	Helps users understand the factors affecting HDI.
4	Feature Importance Analysis	Increases transparency and supports decision-making.
5	Interactive Dashboard	Improves visualization and user experience.

STEP-1

Team Gathering, Collaboration and Select the Problem Statement

TEAM COLLABORATION

Project Name

AI-Powered Human Development Index (HDI) Prediction and Recommendation System

Problem Statement

The Human Development Index (HDI) is a statistical composite index of life expectancy, education, and per capita income indicators, which are used to rank countries into four tiers: Very High, High, Medium, and Low of human development. Current approaches mainly predict the HDI or its category but do not explain the reasons behind the prediction or suggest improvements needed to achieve a higher level of human development.

Our Goal

Predict HDI accurately and provide actionable insights and recommendations to improve human development.

STEP-2

Brainstorm, Idea Listing and Grouping

1

HDI Prediction

Use machine learning to predict HDI score and classify countries into Very High, High, Medium, and Low.

2

Explain Prediction

Identify which indicator (Health, Education, or Income) has the greatest impact on the predicted HDI.

3

Improvement Recommendation

Suggest improvements in life expectancy, education, or income to reach the next HDI category.

4

Feature Importance Analysis

Show the contribution of each feature to help users understand how the prediction was made.

5

Interactive Dashboard

Visualize HDI score, category, comparisons, and recommendations using charts and graphs.

STEP-3

Idea Prioritization

Priority	Idea	Impact	Feasibility	Rationale
★★★★★	Improvement Recommendation	Very High	Medium	Provides actionable guidance and helps users improve HDI to the next level.
★★★★★	HDI Prediction	High	High	Core requirement of the project.
★★★★★	Explain Prediction	High	High	Helps users understand the factors affecting HDI.
★★★★★	Feature Importance Analysis	High	Medium	Increases transparency and builds trust in the model.
★★★★★	Interactive Dashboard	Medium	High	Improves user experience and presentation.